

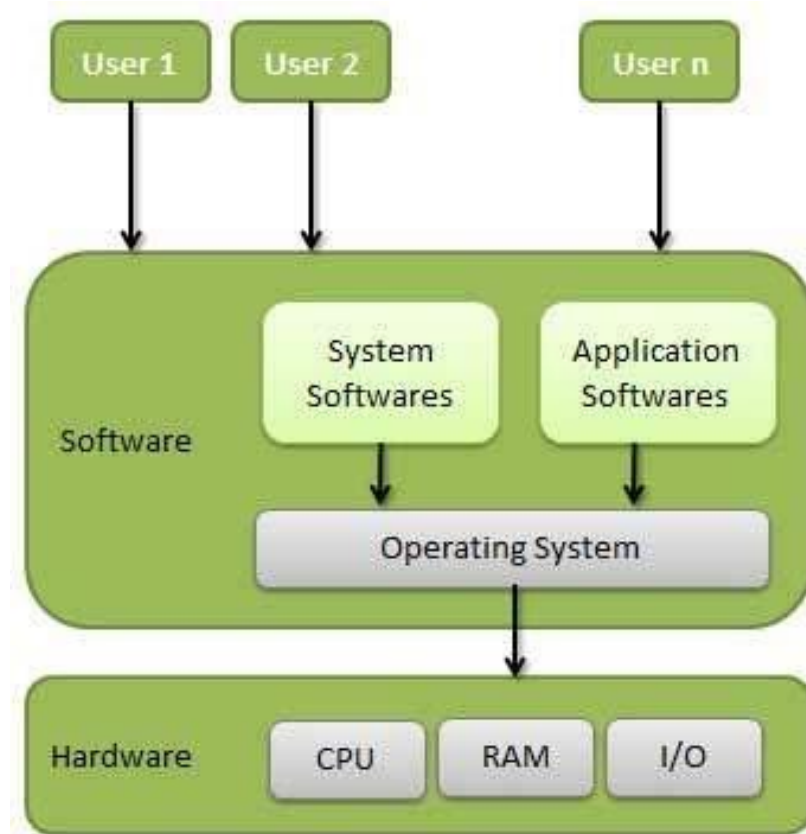
Operating System: Introductory Class

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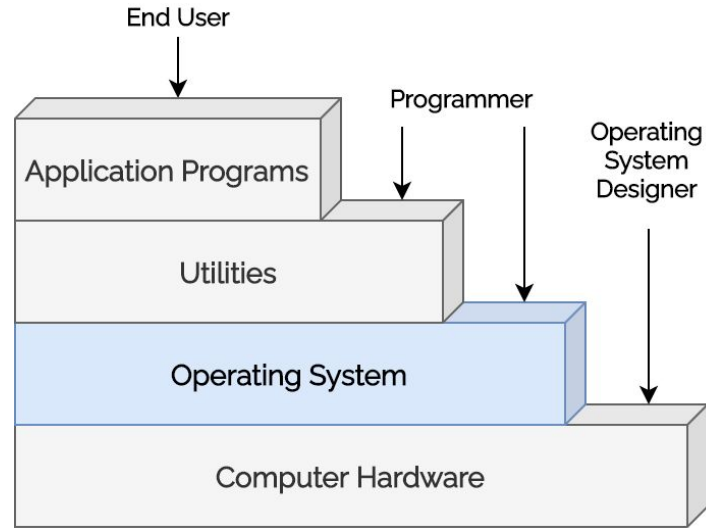
Operating System

- An Operating System (OS) is an interface between a computer user and computer hardware.
- An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.
- Basic tasks or functions
 - Memory Management
 - Processor Management
 - Device Management
 - File Management
 - Security
 - Control over system performance
 - Job accounting
 - Error detecting aids
 - Coordination between other software and users



Objectives

- **Convenience:** An OS makes a computer more convenient to use.
- **Efficiency:** An OS allows the computer system resources to be used in an efficient manner.
- **Ability to evolve:** An OS should be constructed in such a way as to permit the effective development, testing, and introduction of new system functions without interfering with service.
- To hide the details of the hardware resources from the users.
- To keep track of who is using which resource, granting resource requests, and mediating conflicting requests from different programs and users.
- To provide efficient and fair sharing of resources among users and programs.
- To act as an intermediary between the hardware and its users, making it easier for the users to access and use other resources.



Basic Functions: Memory Management

- Management of Primary Memory or Main Memory.
- Main memory provides a fast storage that can be accessed directly by the CPU. **For a program to be executed, it must in the main memory.**
- Keeps tracks of primary memory, i.e., what part of it are in use by whom, what part are not in use.
- In multiprogramming, the OS decides which process will get memory when and how much.
- Allocates the memory when a process requests it to do so.
- De-allocates the memory when a process no longer needs it or has been terminated.

Basic Functions: Processor Management

In multiprogramming environment, the OS decides which process gets the processor when and for how much time. This function is called process scheduling.

- Keeps tracks of processor and status of process. The program responsible for this task is known as traffic controller.
- Allocates the processor (CPU) to a process.
- De-allocates processor when a process is no longer required.

Basic Functions: Device Management

- Keeps tracks of all devices. Program responsible for this task is known as the I/O controller.
- Decides which process gets the device when and for how much time.
- Allocates the device in the efficient way.
- De-allocates devices.

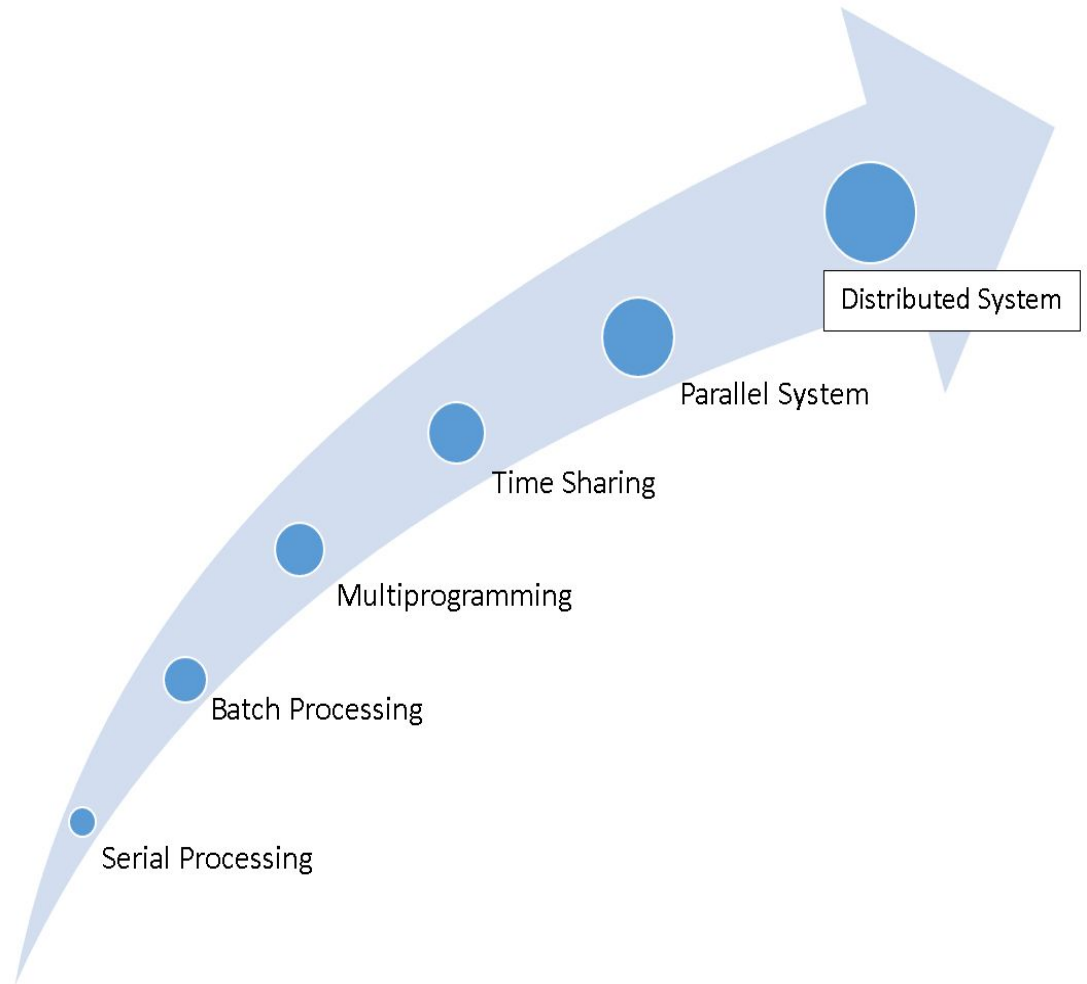
Basic Functions: Others

- **File Management** – Allocates and de-allocates the resources and decides who gets the resources.
- **Security** – Prevents unauthorized access to programs and data by means of passwords and other similar techniques.
- **Job Accounting** – Keeps track of time and resources used by various jobs and/or users.
- **Control Over System Performance** – Records delays between the request for a service and from the system.
- **Interaction with the Operators** – Interaction may take place via the console of the computer in the form of instructions. The Operating System acknowledges the same, does the corresponding action, and informs the operation by a display screen.
- **Error-detecting Aids** – Production of dumps, traces, error messages, and other debugging and error-detecting methods.
- **Coordination Between Other Software and Users** – Coordination and assignment of compilers, interpreters, assemblers, and other software to the various users of the computer systems.

Evolution of OS - Maintainability

- **hardware upgrades or new types of hardware:** With hardware technologies development, the OS also needs to upgrade so as to utilize the new mechanisms introduced by new hardware.
 - For example, Pentium IV extended instruction set of Pentium III for multimedia applications and internet transmission. An OS designed for the previous versions of Intel x86 series will have to be upgraded to be able to accommodate these new instructions.
- **new services:** An OS may also expand to include more services in response to user demand.
- **fixes:** No software is perfect, and any program may contain more or less bugs or defects, thus fixes should be made from time to time. Microsoft Windows is a vivid example of this kind.

Evaluation of Operating System



Evaluation: Serial processing

The earliest computer system has no OS at all, and is characterized as serial processing because users have to reserve time slots in advance, and during the allotted period, they occupy the computer exclusively (Circa 1940-1950)

- Users may finish their tasks earlier than you have expected, and unfortunately the rest time is simply wasted. Or they may run into problems, cannot finish in the allotted time.
- In such systems, programs are presented by cards. Each card has several locations on it, where there may be a hole or not, respectively indicating 0 or 1. Programs are loaded into memory via a card reader. With no OS available, to compile their programs, users have to manually load the compiler program first with the user program as input.

Evaluation: Batch Processing

The central idea is the use of a piece of software known as the monitor. With it, users don't have direct access to the computer systems any longer; instead, the operator of the system collects the user programs and batches them together sequentially for use by the monitor (Circa mid 1950).

- The monitor processed each job in the order it was loaded. When one job finished the monitor ran the next job in line from the batch until all jobs completed.
- Despite an improvement over Serial Processing, Simple Batch Processing was slow and consumed large amounts of processing time.

\$JOB... \$FTN ... FORTRAN
instructions ... \$RUN ... Data ... \$END

Evaluation: Batch Processing

To assist in this, a special language is used, called job control language or JCL (At that time, each user program was called a job), which provides instructions to the monitor.

- \$JOB indicates the beginning of a job.
- \$FTN tells the monitor to load the FORTRAN compiler, which generates object code from the FORTRAN source program. The monitor regains control after the compile operation.
- \$RUN makes control transferred from the monitor to the current program, which works on the data following until a successful or unsuccessful completion.
- The monitor then may move on to another job. To much extent, JCL instructions are similar to the command sequence in a DOS batch file or a UNIX shell file. The latters tell the operating systems to run multiple commands automatically.

Evaluation: Multiprogramming / Multitasking

Even with the automatic job processing by a monitor, the processor is still often idle. The problem is actually what we have discussed before regarding programmed I/O. That is a program may have to wait for I/O operation to finish and thus leads to the processor's idling. The solution is to run multiple programs concurrently

- With multiprogramming, the utilization of processor is greatly improved
- To run multiple programs concurrently, the memory should be organized properly so that each program has its own space and does not invade others
- At some moment, there may be more than programs ready to run. Thus some form of scheduling is needed to obtain better performance.

Evaluation: Time Sharing System

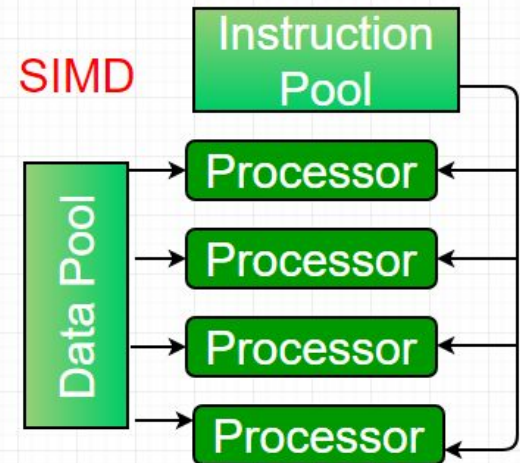
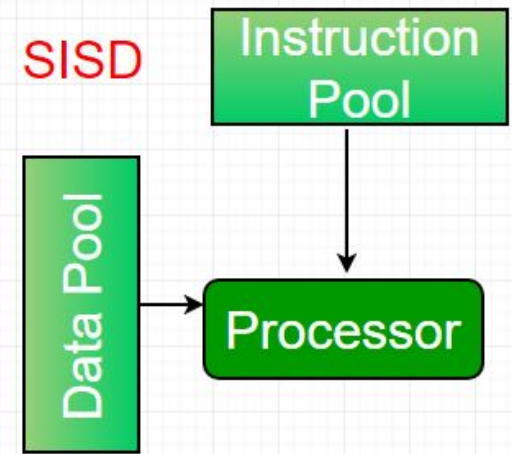
- An extension of Multitasking.
- That is those jobs that come late in the batch job list won't get chance to run until the jobs before them have completed
- In such a system, multiple users simultaneously access the system through terminals, with the operating system interleaving the execution of each user program in a short burst of computation.
 - For example, suppose a computer system may have at most 10 users at the same time, and the human reaction time is 200 ms. Then we may assign $200/10 = 20\text{ms}$ CPU time to the user programs one by one in a cyclic manner, thus each user will be responded within the human reaction time so that the computer system seems to service the user program itself.
- Advantages
 - Provides the advantage of quick response.
 - Avoids duplication of software.
 - Reduces CPU idle time.
- Disadvantages
 - Problem of reliability.
 - Question of security and integrity of user programs and data.
 - Problem of data communication.

Evaluation: Parallel System

- Such systems are multiprocessor systems also known as tightly coupled systems.
- Parallel systems deal with the simultaneous use of multiple computer resources that can include a single computer with multiple processors, a number of computers connected by a network to form a parallel processing cluster or a combination of both.
- Flynn has classified computer systems into four types based on parallelism in the instructions and in the data streams.
 1. Single Instruction stream, single data stream
 2. Single Instruction stream, multiple data stream
 3. Multiple Instruction stream, single data stream
 4. Multiple Instruction stream, multiple data stream

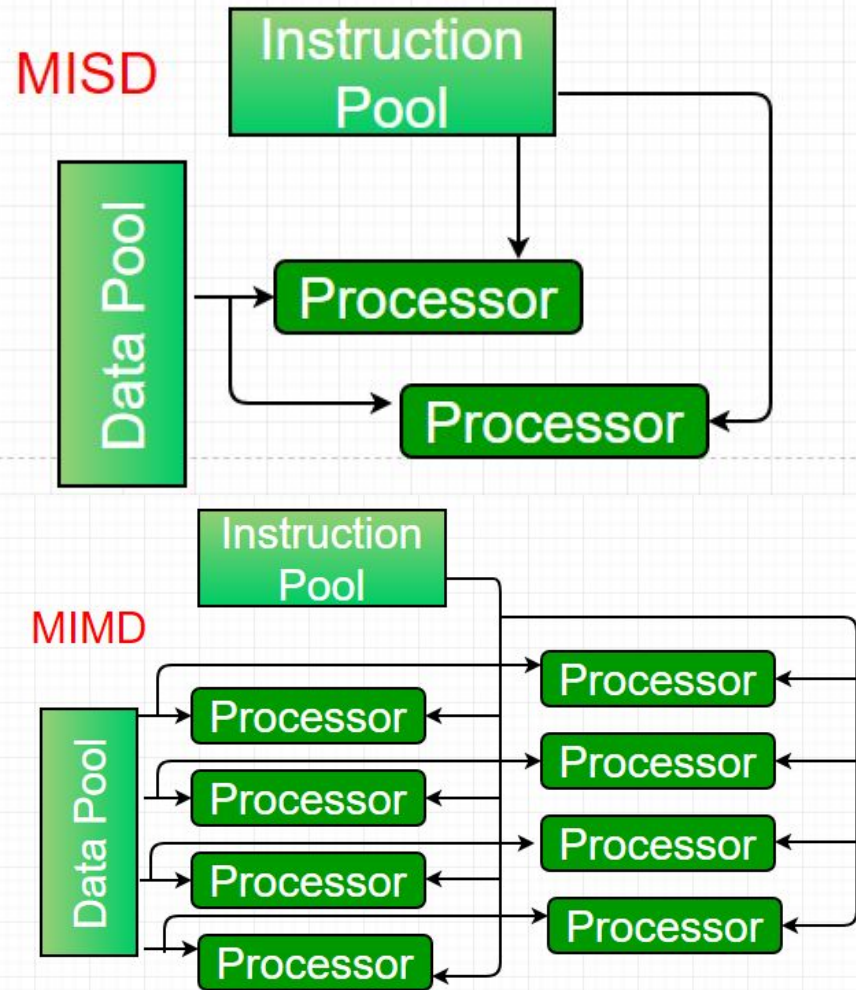
Flynn's taxonomy:Recap

- **Single-instruction, single-data (SISD) systems** - A uniprocessor machine which is capable of executing a single instruction, operating on a single data stream. In SISD, machine instructions are processed in a sequential manner and computers adopting this model are popularly called sequential computers.
- **Single-instruction, multiple-data (SIMD) systems** - A multiprocessor machine capable of executing the same instruction on all the CPUs but operating on different data streams. Machines based on an SIMD model are well suited to scientific computing since they involve lots of vector and matrix operations.



Flynn's taxonomy:Recap

- **Multiple-instruction, single-data (MISD) systems** - a multiprocessor machine capable of executing different instructions on different Processing Elements but all of them operating on the same dataset
 - Example $Z = \sin(x) + \cos(x) + \tan(x)$
- **Multiple-instruction, multiple-data (MIMD) systems** - a multiprocessor machine which is capable of executing multiple instructions on multiple data sets. Each Processing Elements (PE) in the MIMD model has separate instruction and data streams. Unlike SIMD and MISD machines, PEs in MIMD machines work asynchronously.



Evaluation: Distributed System

- Distributed systems use multiple central processors to serve multiple real-time applications and multiple users. Data processing jobs are distributed among the processors accordingly.
- The processors communicate with one another through various communication lines
- Advantages
 - With resource sharing facility, a user at one site may be able to use the resources available at another.
 - Speedup the exchange of data with one another via electronic mail.
 - If one site fails in a distributed system, the remaining sites can potentially continue operating.
 - Better service to the customers.
 - Reduction of the load on the host computer.
 - Reduction of delays in data processing.
- A Network Operating System runs on a server and provides the server the capability to manage data, users, groups, security, applications, and other networking functions. The primary purpose of the network operating system is to allow shared file and printer access among multiple computers in a network